ANAS BENBAKI

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Computer Science Engineering Student | Looking for a 3- to 4-month Internship

EDUCATION

Bordeaux Graduate School of Engineering (ENSEIRB-MATMECA) Bordeaux, France
Fourth year of higher education (2nd year of engineering cycle). 2024 – 2027
5-year engineering degree expected in 2027.
- Intensive coursework in programming, algorithms, probability, statistics, and data science, with applications to AI and quantitative analysis.

LM6E BEN GUERIR, MOROCCO
2-year intensive program in Math and Physics in preparation for the selective entrance examination to French engineering schools. 2022 - 2024
Completed rigorous curriculum covering advanced mathematics and physics.
Awarded scholarship after ranking 1st regionally.

EXPERIENCE

Cap & Pro France Bordeaux, France
Full-stack Intern – Web Development & IVR Integration June 2025 – July 2025
- Designed & deployed a user-centered hotline web platform to improve access to employment assistance for people with disabilities.
- Integrated an IVR system using telephony APIs (call routing, menus, and data collection) to streamline user journeys and reduce manual handling.

Mercor AI San Francisco, US
Mathematics Tutor for AI Models Jan 2025 – May 2025
- Evaluated and corrected AI-generated solutions to advanced mathematics Olympiad problems (algebra, geometry, combinatorics, number theory), providing detailed feedback to improve model reasoning.
- Contributed to RLHF training pipeline by identifying logical errors, verifying proofs, and suggesting alternative solution approaches.

PROJECTS

Reinforcement Learning for Board Game Strategy Optimization (Monopoly)
- Developed a reinforcement learning agent to discover optimal strategies for Monopoly through self-play against rule-based opponents.
- Implemented ϵ-greedy multi-armed bandit algorithm with sample-average Q-value updates to balance exploration and exploitation in decision-making.

Coursidor - Graph-Based Strategy Game
- Collaborated with a 4 teammates to develop a modular game platform in **C** featuring a server-client architecture with dynamically loaded player strategies.
- Implementing graph algorithms (Dijkstra, Minimax with alpha-beta pruning) and sparse matrix representations (GSL CSR/COO) for hexagonal board modeling with 3 topology variants.

Space Shooter Game - Actor-Based Functional Programming
- Co-developed a **TypeScript** shoot'em up game using actor-based architecture and functional programming principles.
- Implementing collision detection algorithms, immutable state management with time-travel mechanics and a terminal Ul with terminal-kit for real-time gameplay visualization.

Aerodynamics Modeling: Airfoil Analysis & Pressure Distribution
- Developed a computational model for airfoil geometry using cubic spline interpolation and adaptive numerical integration (Simpson's rule) to generate smooth aerodynamic profiles and compute curve lengths from discrete data points.
- Applied Bernoulli's equation under laminar flow assumptions to derive and visualize pressure distribution maps along the airfoil surface for lift generation analysis.

ODE Solvers & Dynamical Systems Analysis
- Implemented numerical ODE solvers (Euler, Runge-Kutta methods) in Python to simulate continuous dynamical systems and analyze nonlinear differential equations.
- Built visualization tools (vector fields, phase portraits) to perform qualitative analysis of system dynamics, including equilibrium classification and stability behavior.

Extracurricular Activities

Mathematics Olympiad Training for IMO Team Selection Test: Participated in multiple Mathematics Olympiad Training Sessions organized by the Ministry of Education and the Math & Maroc Association. Focusing on combinatorics, arithmetic, algebra, graph theory, and geometry.
Private Mathematics Tutor: Provided one-on-one coaching to university and high school students in advanced courses like **Linear Algebra, Calculus, & Differential Equations**.

SKILLS

Programming Languages: C, C++, Java, JavaScript, Python, PHP
Technologies & Tools: Bash, Git, Linux/Unix, SQL, SSH
Areas of Expertise: Applied Mathematics, Algorithms, Data Analysis, Machine Learning, Python Programming
Languages: English: Fluent (IELTS 8.0) | French: Fluent | Arabic: Native